

Graph-Based Molecular Generation via Adversarial Learning with Hierarchical Discriminators and Reward-Driven Objectives

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Abstract—Recent advances in deep-generative models have allowed the generation of chemically valid and novel molecules with desirable properties for further investigation. In this work, we provide such a model, integrating generative adversarial networks (GANs), hierarchical graph discrimination, and reinforcement learning. Our graph-based generator generates molecular graphs in the form of adjacency and node feature matrices. To improve robustness and stability during training, we implement a hierarchical differentiable pooling (DIFFPOOL)-based discriminator. This type of discriminator allows our model to learn coarse-grained graph representations during training. Additionally, we introduce a reinforcement learning component in which a policy network receives reward signals based on predicted chemical properties, guiding the generator toward structures with high drug-likeness and other target metrics. In evaluating our approach, we look for validity, uniqueness, novelty, and alignment with drug-likeness metric scores. The results demonstrate improved generation quality and training stability compared to baseline graph GANs, particularly when incorporating DIFFPOOL and reinforcement-guided reward shaping. Our findings suggest that the combination of hierarchical discrimination with reinforcement learning shows a positive trajectory for property-driven molecular design in drug discovery and materials science.

Index Terms—GAN, Hierarchical discriminator, DIFFPOOL, Reinforcement learning, Validity, Uniqueness, Novelty, Drug-likeness

Nomenclature

$A \in \mathbb{R}^{n \times n}$	adjacency matrix
$T = [t_1, t_2, \dots, t_n] \in \mathbb{Z}^n$	atomic numbers (node features)
$B \in \mathbb{Z}^{n \times n}$	bond order matrix (optional)
$G = (V, E)$	molecular graph with vertices V and edges E

I. INTRODUCTION

In our research, we used the QM9 dataset to train a GAN generator working alongside a hierarchical discriminator to generate chemically valid and novel molecules for scientific

study. We used two different versions of the hierarchical discriminator, one with DIFFPOOL, and one without. In association with the generator and discriminator, we also implement a reward policy network to encourage the model’s generation. The dataset we used, QM9, contained the following variables: mol_id, smiles, A, B, C, mu, alpha, homo, lumo, gap, r2, zpve, u0, u298, h298, g298, cv, u0_atom, u298_atom, h298_atom, g298_atom, Morgan_fingerprint, adj_matrix, node_features, and node_count. The variables that are significant for this work are adj_matrix, node_features, and node_count. These variables define the molecules that are fed into the generator during training. The adjacency matrix, adj_matrix, contains information on which bonds are present inside the molecule. The node features, node_features, contain information on which atoms are present within the molecule. In this study, we analyze the validity, novelty, uniqueness, and drug-likeness of molecules produced by the generator. The validity is based on whether or not the generated molecule is chemically valid, uniqueness is based on similarity of the generated molecule to previous molecules, novelty is determined by whether or not a similar molecule is already present within the training data, and drug-likeness is concerned with the similarity of the molecule generated to common drug-like molecules.

II. LITERARY REVIEW

III. METHODS

A. Adjacency Matrix to Molecule

1) *Vertex Set (Atoms)*: Each atom is represented by a vertex:

$$V = \{v_i = \text{Atom}(t_i) | i = 1, 2, \dots, n\} \quad (1)$$

Optionally, if chirality tags are given:

$$R_{\text{validity}}(M) = \begin{cases} -0.5, & \text{if } M \text{ is None or fails sanitization} \\ \max(0, 5 - |\sum_{v_i \in V} (\text{Valence}_{\max}(t_i) - \deg(v_i))|), & \text{otherwise} \end{cases} \quad (10)$$

$$\text{Chiral}(v_i) = c_i \text{ for } i \in C \subseteq \{1, \dots, n\} \quad (2)$$

2) *Edge Set (Bonds)*: For all $i < j$:

$$\text{If } A_{ij} > \tau : \quad o_{ij} = \begin{cases} B_{ij}, & \text{if } B_{ij} > 0 \\ 1, & \text{otherwise} \end{cases} \quad (3)$$

$$E \ni e_{ij} = (v_i, v_j, \beta(o_{ij}))$$

Where $\beta(o)$ maps bond order to bond type:

$$\beta(o) = \begin{cases} \text{SINGLE}, & o = 1 \\ \text{DOUBLE}, & o = 2 \\ \text{TRIPLE}, & o = 3 \\ \text{AROMATIC}, & o = 4 \\ \text{undefined}, & \text{otherwise} \end{cases} \quad (4)$$

3) *Isolated Hydrogen Handling*: Let:

$$H = \{i | t_i = 1 \wedge \deg(v_i) = 0\} \quad (5)$$

For each $h \in H$:

$$j^* = \arg \max_{\substack{j \notin H \\ j \neq h}} A_{hj} \quad (6)$$

If $\deg(v_{j^*}) < \text{Valence}_{\max} t_{j^*}$, then:

$$E \ni e_{hj^*} = (v_h, v_{j^*}, \text{SINGLE}) \quad (7)$$

Else:

$$V \leftarrow V \setminus \{v_h\} \quad (8)$$

4) *Molecule*: The final molecule is:

$$M = \text{Sanitize}(G = (V, E)) \quad (9)$$

B. Reward Policy Network

1) *Validity Reward*: Let M be a candidate molecule derived from (A, T) . Let $\text{Valence}_{\max}(t_i)$ be the maximum valence for atom type t_i , and let $\deg(v_i)$ be the current explicit valence (degree) of atom $v_i \in V$. See Eq. ?? for the validity reward function.

2) *Stray Hydrogen Reward*: Let $H \subset V$ be the set of hydrogen atoms, and let $\deg(v_i)$ be the degree of atom v_i .

$$R_{\text{stray}}(M) = \begin{cases} -0.5, & \text{if } M \text{ is None or } \exists v_i \in H \text{ such that } \deg(v_i) = 0 \\ 0.5, & \text{otherwise} \end{cases} \quad (11)$$

3) *Uniqueness Reward*: Let $\mathcal{G} = \{S_1, S_2, \dots, S_n\}$ be the list of SMILES strings for previously generated molecules, and let $S(M)$ be the SMILES of the current molecule M . Let $\text{count}_{\mathcal{G}}(S(M))$ be the number of times $S(M)$ appears in $\mathcal{G}$.

$$R_{\text{unique}}(M, \mathcal{G}) = \begin{cases} -0.3, & \text{if } R_{\text{validity}}(M) = 0 \\ \max\left(0, \min\left(1, 1 - \frac{|G|+1}{\text{count}_{\mathcal{G}}(S(M))}\right)\right), & \text{otherwise} \end{cases} \quad (12)$$

4) *Novelty Reward*: Let $\mathcal{T} \subset \mathbb{S}$ be the set of training SMILES strings.

$$R_{\text{novelty}}(M, \mathcal{T}, \mathcal{G}) = \begin{cases} -0.3, & \text{if } R_{\text{validity}}(M) = 0 \\ -2.5, & \text{if } R_{\text{unique}}(M, \mathcal{G}) < 1 \\ -0.7, & \text{if } S(M) \in \mathcal{T} \\ 2.3, & \text{otherwise} \end{cases} \quad (13)$$

IV. EASE OF USE

A. Maintaining the Integrity of the Specifications

The IEEEtran class file is used to format your paper and style the text. All margins, column widths, line spaces, and text fonts are prescribed; please do not alter them. You may note peculiarities. For example, the head margin measures proportionately more than is customary. This measurement and others are deliberate, using specifications that anticipate your paper as one part of the entire proceedings, and not as an independent document. Please do not revise any of the current designations.

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Define abbreviations and acronyms the first time they are used in the text, even after they have been defined in the abstract. Abbreviations such as IEEE, SI, MKS, CGS, ac, dc, and rms do not have to be defined. Do not use abbreviations in the title or heads unless they are unavoidable.

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C. Equations

Number equations consecutively. To make your equations more compact, you may use the solidus (/), the exp function, or appropriate exponents. Italicize Roman symbols for quantities and variables, but not Greek symbols. Use a long dash rather than a hyphen for a minus sign. Punctuate equations with commas or periods when they are part of a sentence, as in:

$$a + b = \gamma \quad (14)$$

Be sure that the symbols in your equation have been defined before or immediately following the equation. Use “(??)”, not “Eq. (??)” or “equation (??)”, except at the beginning of a sentence: “Equation (??) is . . .”

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Please use “soft” (e.g., `\eqref{Eq}`) cross references instead of “hard” references (e.g., (1)). That will make it possible to combine sections, add equations, or change the order of figures or citations without having to go through the file line by line.

Please don’t use the `{eqnarray}` equation environment. Use `{align}` or `{IEEEeqnarray}` instead. The `{eqnarray}` environment leaves unsightly spaces around relation symbols.

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E. Some Common Mistakes

- The word “data” is plural, not singular.
- The subscript for the permeability of vacuum μ_0 , and other common scientific constants, is zero with subscript formatting, not a lowercase letter “o”.

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- There is no period after the “et” in the Latin abbreviation “et al.”.
- The abbreviation “i.e.” means “that is”, and the abbreviation “e.g.” means “for example”.

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Headings, or heads, are organizational devices that guide the reader through your paper. There are two types: component heads and text heads.

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Text heads organize the topics on a relational, hierarchical basis. For example, the paper title is the primary text head because all subsequent material relates and elaborates on this one topic. If there are two or more sub-topics, the next level head (uppercase Roman numerals) should be used and, conversely, if there are not at least two sub-topics, then no subheads should be introduced.

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a) *Positioning Figures and Tables:* Place figures and tables at the top and bottom of columns. Avoid placing them in the middle of columns. Large figures and tables may span across both columns. Figure captions should be below the figures; table heads should appear above the tables. Insert figures and tables after they are cited in the text. Use the abbreviation “Fig. ??”, even at the beginning of a sentence.

TABLE I: Table Type Styles

Table Head	Table Column Head		
	Table column subhead	Subhead	Subhead
copy	More table copy ^a		

^aSample of a Table footnote.

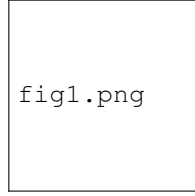


Fig. 1: Example of a figure caption.

Figure Labels: Use 8 point Times New Roman for Figure labels. Use words rather than symbols or abbreviations when writing Figure axis labels to avoid confusing the reader. As an example, write the quantity “Magnetization”, or “Magnetization, M”, not just “M”. If including units in the label, present them within parentheses. Do not label axes only with units. In the example, write “Magnetization (A/m)” or “Magnetization {A[m(1)]}”, not just “A/m”. Do not label axes with a ratio of quantities and units. For example, write “Temperature (K)”, not “Temperature/K”.

ACKNOWLEDGMENT

The preferred spelling of the word “acknowledgment” in America is without an “e” after the “g”. Avoid the stilted expression “one of us (R. B. G.) thanks . . .”. Instead, try “R. B. G. thanks . . .”. Put sponsor acknowledgments in the unnumbered footnote on the first page.

REFERENCES

Please number citations consecutively within brackets [?]. The sentence punctuation follows the bracket [?]. Refer simply to the reference number, as in [?]¹—do not use “Ref. [?]” or “reference [?]” except at the beginning of a sentence: “Reference [?] was the first . . .”

Number footnotes separately in superscripts. Place the actual footnote at the bottom of the column in which it was cited. Do not put footnotes in the abstract or reference list. Use letters for table footnotes.

Unless there are six authors or more give all authors’ names; do not use “et al.”. Papers that have not been published, even if they have been submitted for publication, should be cited as “unpublished” [?]. Papers that have been accepted for publication should be cited as “in press” [?]. Capitalize only the first word in a paper title, except for proper nouns and element symbols.

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